

PC Build Store Management System

System Architecture & ER Diagram

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1. Architecture Overview

The app has 3 layers:

GUI Layer (Swing panels)
Business Logic Layer (DAOs)
Database Layer (MySQL via JDBC)

DashboardGUI is the main frame. It uses CardLayout to switch between modules. Each module has its own panel that gets shown or hidden.

2. Entity Relationship Diagram

[ER diagram image not found]

3. Table Details

builds (184 rows)

- build_id: auto increment primary key
- cpu_model: what CPU is in the build
- cpu_type: Intel or AMD
- gpu_model: what GPU is in the build
- gpu_type: Nvidia or AMD
- ram, storage, psu: hardware specs
- total_price: price in PKR (50k-500k)
- performance_score: 0-120 scale

gpu_options (59 rows)

- gpu_id: auto increment primary key
- brand: Nvidia or AMD
- name: GPU model name
- for_budget: which budget tier this GPU is for
- price_increase: how much the GPU costs
- performance_increase: how much performance it adds

bills (variable)

- bill_id: auto increment primary key
- build_id: which build was bought
- final_cpu, final_gpu: from the purchased build
- final_price: price paid

- final_score: performance score
- purchase_date: when the purchase happened

4. SQL Schema

```
CREATE TABLE builds (  
  build_id INT(11) NOT NULL AUTO_INCREMENT,  
  cpu_model VARCHAR(100) NOT NULL,  
  cpu_type VARCHAR(10) NOT NULL,  
  gpu_model VARCHAR(100) NOT NULL,  
  gpu_type VARCHAR(10) NOT NULL,  
  ram VARCHAR(50) NOT NULL,  
  storage VARCHAR(50) NOT NULL,  
  psu VARCHAR(50) NOT NULL,  
  total_price INT(11) NOT NULL,  
  performance_score INT(11) NOT NULL,  
  PRIMARY KEY (build_id)  
);
```

```
CREATE TABLE gpu_options (  
  gpu_id INT(11) NOT NULL AUTO_INCREMENT,  
  brand VARCHAR(10) NOT NULL,  
  name VARCHAR(100) NOT NULL,  
  for_budget INT(11) NOT NULL,  
  price_increase INT(11) NOT NULL,  
  performance_increase INT(11) NOT NULL,  
  PRIMARY KEY (gpu_id)  
);
```

```
CREATE TABLE bills (  
  bill_id INT(11) NOT NULL AUTO_INCREMENT,  
  build_id INT(11) NOT NULL,  
  final_cpu VARCHAR(100) NOT NULL,  
  final_gpu VARCHAR(100) NOT NULL,  
  final_price INT(11) NOT NULL,  
  final_score INT(11) NOT NULL,  
  purchase_date DATETIME DEFAULT CURRENT_TIMESTAMP,  
  PRIMARY KEY (bill_id),  
  FOREIGN KEY (build_id) REFERENCES builds(build_id)  
);
```

5. Connection

```
jdbc:mysql://localhost:3306/pc_build_store  
?useSSL=false&serverTimezone=UTC  
&allowPublicKeyRetrieval=true  
  
Username: root  
Password: (empty)
```